

Collaboration Policy for Team Data Science Projects

1. Purpose

The purpose of this collaboration policy is to ensure that students working in groups on the **Team Data Science Project** engage in productive, fair, and professional teamwork. Data science projects are inherently collaborative and require coordination between team members with different skills, including data engineering, modeling, domain analysis, and visualization.

This policy establishes guidelines to promote **accountability, transparency, and effective collaboration**, while ensuring that each student contributes meaningfully to the project.

2. Team Formation

Students will work in **teams of three to five members**. Each team should aim to include members with complementary skills such as programming, data analysis, machine learning, domain knowledge, and data visualization.

Each team must assign the following **core roles**:

- **Data Engineer:** Responsible for data acquisition, preprocessing, and data pipeline management.

- **Machine Learning Specialist:** Responsible for model development, training, and evaluation.
- **Domain Lead:** Responsible for defining the problem context and ensuring alignment with the project objectives.
- **Visualization & Storytelling Lead:** Responsible for presenting insights through dashboards, visualizations, and reports.
- **Project Coordinator:** Responsible for task coordination, meeting organization, documentation, and ensuring deadlines are met.

Although each member has a primary role, **all team members are expected to understand the entire data science pipeline and contribute collaboratively.** For teams with fewer than five members, a team member may perform more than one role.

3. Individual Responsibilities

Each student must:

- Actively participate in **team meetings and discussions.**
- Contribute to **data collection, analysis, modeling, and reporting tasks.**
- Demonstrate understanding of the **entire project workflow**, not just their assigned role.
- Complete assigned tasks **on time and with academic integrity.**

- Maintain respectful and professional communication with teammates.

Students should maintain **shared documentation** showing their contributions, including code commits, data processing steps, and analysis outputs.

4. Communication and Meetings

To maintain effective collaboration, each team must:

- Hold **regular meetings** (at least once per week).
- Document the meeting minutes, including assigned tasks and deadlines.
- Use collaborative tools such as:
 - GitHub / GitLab for version control
 - Google Drive / OneDrive for shared documents
 - Slack / Microsoft Teams / WhatsApp for communication
 - Trello / Notion (or similar tools) for task management

Teams must ensure **transparent communication**, where all members are aware of project progress and responsibilities. You may choose different tools not listed above that fit your needs or are convenient.

5. Task Distribution and Work Tracking

Each project should include a **clear task allocation plan**.

Teams should:

- Divide tasks based on expertise and interest.
- Maintain a **task tracker** showing who is responsible for each task.
- Use version control tools to record code contributions.

Examples of tasks include:

- Data collection and cleaning
- Feature engineering
- Model development
- Model evaluation
- Visualization and dashboard creation
- Writing the final report

This ensures that the **workload is distributed fairly among all team members**.

6. Academic Integrity and Ethical Collaboration

Students must follow principles of **academic honesty and ethical data use**.

Students must:

- Avoid plagiarism in code, analysis, or written reports.
- Cite external datasets, libraries, and research sources.
- Respect privacy and data protection regulations when using real datasets.
- Ensure that all team members receive proper credit for their contributions.

Collaborative work must reflect **genuine participation from all group members**.

7. Conflict Resolution

Team disagreements may arise during the project. When conflicts occur, students should follow these steps:

1. Attempt to resolve the issue through **open discussion within the team**.
2. Document the issue and proposed solutions.
3. If unresolved, report the issue to the **course instructor or lecturer**.

The instructor may intervene by:

- Redistributing tasks
- Mediating discussions
- Evaluating individual contributions

Persistent lack of cooperation may lead to **individual grading adjustments**.

8. Participation and Contribution Evaluation

Teamwork will be evaluated based on both **group performance and individual contributions**.

Assessment may include:

- Participation in meetings and project discussions
- Contribution to coding, modeling, and analysis
- Quality of assigned tasks
- Peer evaluation within the team
- Participation in presentations and demonstrations

Students who do not actively contribute may receive **lower individual grades** even if the group performs well.

9. Deliverables and Documentation

Each team must submit the following deliverables:

- Project proposal (if the topic is different from the one proposed by the lecturer)
- Data exploration and analysis report
- Machine learning models and code repository
- Visualizations and dashboards
- Final written report
- Final presentation and demonstration

All deliverables should be **shared and documented collaboratively**.

10. Professional Conduct

Students are expected to demonstrate professional behavior similar to that expected in real-world data science teams.

This includes:

- Respectful communication
- Responsibility for assigned tasks
- Openness to feedback
- Collaboration and knowledge sharing
- Commitment to deadlines and project quality

Effective collaboration is essential for successful data science projects. By following this collaboration policy, students will gain valuable experience in **team-based problem solving, interdisciplinary communication, and real-world data science workflows**.

These skills are critical for preparing students to work in **professional data science teams in industry, research, and public organizations**.